Rumor spreading by uniform push on the complete graph: a formalization-oriented proof of the $O(\log n)$ bound

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Abstract

We give a complete and elementary proof that the uniform *push* protocol on the complete graph K_n, started from a single informed node, informs all n nodes within $O(\log n)$ rounds with high probability. The proof is deliberately optimized for formalization in a proof assistant (Lean 4 + Mathlib): the probability space is finite and uniform; the only analytic ingredients are Bernoulli's inequality and Markov's inequality; concentration in the growth phase is obtained from a one-line exponential-moment induction for adaptive Bernoulli trials, avoiding both Chernoff bounds for sums of independent variables and stochastic-domination couplings. No use is made of the exponential function, of continuous probability, or of martingale theory. Constants are deliberately crude.

This argument has been *fully formalized* (no `sorry`) in the accompanying Lean development (`RumorSpread/`); the main theorem is `RumorPush.push_informs_all_whp`.

1 The model

Fix $n \geq 2$ and let $V = \{0, 1, \ldots, n-1\}$ be the vertex set of the complete graph K_n.

Definition 1.1 (Round randomness). *A* round configuration *is a function $\omega \colon V \to V$ with $\omega(v) \neq v$ for all v. Let R_n denote the (finite) set of round configurations; it carries the uniform probability measure, i.e. every node picks a uniformly random* target *among the other $n-1$ nodes, independently of all other nodes.*

Definition 1.2 (Push step). *For an* informed set *$I \subseteq V$ and a round configuration $\omega \in R_n$, the next informed set is*

$$\operatorname{step}(I, \omega) \;=\; I \cup \omega(I) \;=\; I \cup \{\omega(v) : v \in I\}.$$

(Only informed nodes actually transmit; letting all *nodes draw a target and ignoring the choices of uninformed nodes is an equivalent and formalization-friendlier convention.)*

Definition 1.3 (Process). *Fix a horizon $T \in \mathbb{N}$ and a start vertex $v_0 \in V$. The sample space is*

$$\Omega \;=\; R_n^T \quad \textit{(uniform measure, i.e. i.i.d. uniform rounds).}$$

For $\omega = (\omega_0, \ldots, \omega_{T-1}) \in \Omega$ define $I_0(\omega) = \{v_0\}$ and $I_{t+1}(\omega) = \operatorname{step}(I_t(\omega), \omega_t)$ for $0 \leq t < T$. We write $m_t = |I_t|$ and $U_t = n - m_t$ for the numbers of informed and uninformed nodes. Since Ω is finite, all probabilities are finite ratios and all expectations are finite sums; "conditioning on I_t" means partitioning Ω according to the first t coordinates.

Two trivial but load-bearing facts, both immediate from Definition 1.2:

Lemma 1.4 (Monotonicity). *$I_t(\omega) \subseteq I_{t+1}(\omega)$ for all t and all ω; hence m_t is non-decreasing and U_t is non-increasing in t, pointwise on Ω.*

Lemma 1.5 (At most doubling). *$|I_{t+1}(\omega) \setminus I_t(\omega)| \leq |I_t(\omega)|$, i.e. the number $N_t := m_{t+1} - m_t$ of newly informed nodes in one round satisfies $0 \leq N_t \leq m_t$.*

2 Main theorem

Throughout, log denotes the natural logarithm.

Theorem 2.1 (Push protocol informs K_n in $O(\log n)$ rounds w.h.p.)**.** *Let $n \geq 2$, let $v_0 \in V$, and set*

$$T_1 \;=\; \lceil 117 \log n \rceil + 23, \qquad T_2 \;=\; \lceil 6 \log n \rceil, \qquad T \;=\; T_1 + T_2.$$

Then

$$\mathbb{P}\big[\, I_T = V \,\big] \;\geq\; 1 - \frac{2}{n}.$$

In particular all nodes are informed after $O(\log n)$ rounds with high probability.

Remark 2.2. *The constants are deliberately crude: the proof is organized to minimize the analytic toolkit, not the round count. The sharp answer is $\log_2 n + \log n + o(\log n)$ (Frieze–Grimmett 1985, Pittel 1987). Replacing $2/n$ by $n^{-\alpha}$ for any fixed $\alpha \geq 1$ only changes the constants in T_1, T_2.*

The proof has two phases, glued by Lemma 1.4:

- **Growth phase** (Section 6): while $m_t \leq n/2$, each round multiplies m_t by $\geq 9/8$ with probability $\geq 1/8$; an exponential-moment bound for adaptive trials shows that after T_1 rounds, $m_{T_1} > n/2$ except with probability $\leq 1/n$.
- **Saturation phase** (Section 7): once $m_t \geq n/2$, the expected number of uninformed nodes contracts by a factor $2/3$ per round; after T_2 further rounds Markov's inequality gives $U_T = 0$ except with probability $\leq 1/n$.

3 Elementary inequalities

Everything in this section is deterministic real arithmetic. We write $x := \frac{1}{n-1}$, so $0 < x \leq 1$ for $n \geq 2$.

Lemma 3.1 (Bernoulli)**.** *For all real $y \geq -1$ (we only use $y \in [-1, 1]$) and $m \in \mathbb{N}$: $(1+y)^m \geq 1 + my$.*

Proof. Induction on m; available in Mathlib as `one_add_mul_le_pow`. ☐

Lemma 3.2 (Second-order lower bound, "Bonferroni")**.** *For $0 \leq x \leq 1$ and $m \in \mathbb{N}$:*

$$1 - (1-x)^m \;\geq\; mx - \binom{m}{2} x^2 \;\geq\; mx\Big(1 - \frac{mx}{2}\Big).$$

Proof. Induction on m. The case $m = 0$ is trivial. For the step, using Lemma 3.1 in the form $(1-x)^m \geq 1 - mx$:

$$1-(1-x)^{m+1} = \big(1-(1-x)^m\big) + x(1-x)^m \geq \Big(mx - \tbinom{m}{2}x^2\Big) + x(1-mx) = (m+1)x - \tbinom{m+1}{2}x^2,$$

since $\binom{m}{2} + m = \binom{m+1}{2}$. The second inequality is $\binom{m}{2} \leq m^2/2$. ☐

Lemma 3.3 (Upper bound without e)**.** *For $0 \leq x \leq 1$ and $m \in \mathbb{N}$:*

$$(1-x)^m \;\leq\; \frac{1}{1+mx}.$$

Proof. $(1-x)(1+x) = 1-x^2 \le 1$, so $(1-x)^m(1+x)^m \le 1$, and $(1+x)^m \ge 1+mx > 0$ by Lemma 3.1; divide. $\square$

Lemma 3.4 (Logarithm bounds)**.** *For $x > 0$: $1 - \frac{1}{x} \le \log x \le x - 1$.*

Proof. The right inequality is standard (concavity of log, in Mathlib as `Real.log_le_sub_one_of_pos`); the left follows by applying it to $1/x$. These two bounds, together with Mathlib's numeric bound $\log 2 < 0.6931471808$, are the only facts about log used in the numeric verification of Lemma 6.4; in particular $\log \frac{9}{8} \ge \frac{1}{9}$, $\log \frac{16}{15} \ge \frac{1}{16}$ and $\log \frac{3}{2} \ge \frac{1}{3}$. $\square$

Lemma 3.5 (Markov's inequality, finite uniform version)**.** *Let Ω be a finite nonempty set with the uniform measure, $f \colon \Omega \to \mathbb{R}$ with $f \ge 0$, and $a > 0$. Then $\mathbb{P}[f \ge a] \le \mathbb{E}[f]/a$. In particular, if f takes values in $\mathbb{N}$, then $\mathbb{P}[f \ne 0] \le \mathbb{E}[f]$.*

Lemma 3.6 (Reverse Markov for bounded variables)**.** *Let $f \colon \Omega \to \mathbb{R}$ with $0 \le f \le b$ pointwise, $b > 0$, and let $0 < a < b$. Then*

$$\mathbb{P}[f \ge a] \;\ge\; \frac{\mathbb{E}[f] - a}{b}.$$

Proof. $\mathbb{E}[f] \le a\,\mathbb{P}[f < a] + b\,\mathbb{P}[f \ge a] \le a + b\,\mathbb{P}[f \ge a]$. $\square$

4 One-round estimates

Fix an informed set I with $|I| = m \ge 1$ and let $\omega \in R_n$ be a uniform round configuration. All statements in this section concern this single round (one uniform draw from R_n); by independence of the rounds they apply verbatim to the conditional law of round t given $I_t = I$.

Lemma 4.1 (Contact probability)**.** *For any fixed $u \notin I$,*

$$\mathbb{P}_\omega\big[u \in \mathrm{step}(I,\omega)\big] \;=\; 1 - \Big(1 - \frac{1}{n-1}\Big)^m.$$

Proof. $u \notin \mathrm{step}(I,\omega)$ iff $\omega(v) \ne u$ for every $v \in I$. The coordinates $(\omega(v))_{v\in I}$ are independent and uniform on $V \setminus \{v\}$, and $u \ne v$ for $v \in I$, so each misses u with probability exactly $1 - \frac{1}{n-1}$; multiply over $v \in I$. (Formally: count the functions ω avoiding u on I; the count factorizes coordinate-by-coordinate.) $\square$

Lemma 4.2 (Expected growth while below half)**.** *Let $N(\omega) = |\mathrm{step}(I,\omega)| - m$ be the number of newly informed nodes. If $1 \le m \le n/2$, then*

$$\mathbb{E}_\omega[N] \;\ge\; \frac{m}{4}.$$

Proof. Write $x = \frac{1}{n-1}$. By linearity of expectation over the $n-m$ uninformed nodes and Lemma 4.1, $\mathbb{E}[N] = (n-m)\big(1-(1-x)^m\big)$. Since $m \le n/2$ and $n \ge 2$ we have $mx = \frac{m}{n-1} \le \frac{n}{2(n-1)} \le 1$, so Lemma 3.2 gives $1-(1-x)^m \ge mx(1-\frac{mx}{2}) \ge \frac{mx}{2}$. Using $n - m \ge n/2$:

$$\mathbb{E}[N] \;\ge\; \frac{n}{2}\cdot\frac{m}{2(n-1)} \;=\; \frac{m}{4}\cdot\frac{n}{n-1} \;\ge\; \frac{m}{4}. \qquad \square$$

Lemma 4.3 (A good round has constant probability)**.** *If $1 \le m \le n/2$, then*

$$\mathbb{P}_\omega\Big[\,|\mathrm{step}(I,\omega)| \ge \tfrac{9}{8}\,m\Big] \;\ge\; \frac{1}{8}.$$

Proof. N satisfies $0 \le N \le m$ (Lemma 1.5) and $\mathbb{E}[N] \ge m/4$ (Lemma 4.2). By Lemma 3.6 with $a = m/8$, $b = m$:

$$\mathbb{P}\big[N \ge \tfrac{m}{8}\big] \;\ge\; \frac{m/4 - m/8}{m} = \frac{1}{8}. \qquad \square$$

Lemma 4.4 (Contraction above half)**.** *If $m \ge n/2$, then for every fixed $u \notin I$,*

$$\mathbb{P}_\omega\big[u \notin \mathrm{step}(I,\omega)\big] \;=\; \Big(1 - \frac{1}{n-1}\Big)^m \;\le\; \frac{2}{3},$$

and consequently, with $U' := n - |\mathrm{step}(I,\omega)|$,

$$\mathbb{E}_\omega[U'] \;\le\; \frac{2}{3}\,(n-m).$$

Proof. By Lemma 3.3, $(1-x)^m \le \frac{1}{1+mx}$, and $mx = \frac{m}{n-1} \ge \frac{n}{2(n-1)} \ge \frac{1}{2}$, so $(1-x)^m \le \frac{1}{3/2} = \frac{2}{3}$. Summing over the $n-m$ uninformed nodes gives the expectation bound. $\square$

5 Adaptive Bernoulli trials: an exponential-moment bound

The only "concentration" tool we need. We state it for our concrete finite product space; $\mathcal{F}_t$-measurable simply means "a function of the first t coordinates of $\omega \in \Omega = R_n^T$".

Lemma 5.1 (Exponential moment for adaptive trials)**.** *Let $X_0, \dots, X_{T-1} \colon \Omega \to \{0,1\}$ be such that X_t is a function of $(\omega_0, \dots, \omega_t)$, and suppose there is $p \in [0,1]$ such that for every t and every fixed prefix $(\omega_0, \dots, \omega_{t-1})$,*

$$\mathbb{P}\big[X_t = 1 \,\big|\, \omega_0, \dots, \omega_{t-1}\big] \;\ge\; p.$$

Let $G = \sum_{t<T} X_t$. Then for every $s \in (0,1]$,

$$\mathbb{E}\big[s^G\big] \;\le\; \big(1 - p(1-s)\big)^T.$$

Proof. Induction on T. For $T = 0$ both sides are 1. For the step, condition on the first T coordinates: with $G' = \sum_{t<T} X_t$,

$$\mathbb{E}\big[s^{G'+X_T}\big] = \mathbb{E}\Big[s^{G'} \cdot \mathbb{E}\big[s^{X_T} \mid \omega_0, \dots, \omega_{T-1}\big]\Big],$$

and pointwise $\mathbb{E}[s^{X_T} \mid \cdot] = 1 - q(1-s) \le 1 - p(1-s)$ where $q = \mathbb{P}[X_T = 1 \mid \cdot] \ge p$ and $1 - s \ge 0$. Since $s^{G'} \ge 0$, the induction hypothesis applies to the outer expectation. $\square$

Corollary 5.2 (Lower tail)**.** *In the setting of Lemma 5.1, for every $L \in \mathbb{N}$ and $s \in (0,1]$,*

$$\mathbb{P}[G < L] \;\le\; s^{-L}\,\big(1 - p(1-s)\big)^T.$$

With $p = \frac{1}{8}$ and $s = \frac{1}{2}$: $\mathbb{P}[G < L] \le 2^L \left(\frac{15}{16}\right)^T$.

Proof. On $\{G < L\}$ we have $s^G \ge s^L$ (as $s \le 1$), so $s^L \mathbb{P}[G < L] \le \mathbb{E}[s^G]$; apply Lemma 5.1 and Markov's reasoning directly (no measure theory needed: these are finite sums). $\square$

Remark 5.3. *Lemma 5.1 replaces, in one induction, both the stochastic domination of adaptive trials by a binomial and the Chernoff bound for the binomial lower tail. This is the main simplification that makes the growth phase formalization-friendly.*

6 Growth phase

Definition 6.1 (Good rounds)**.** *For $t < T$ let*

$$X_t \;=\; \mathbf{1}\Big[\, m_{t+1} \ge \tfrac{9}{8}\, m_t \;\; \textit{or} \;\; m_t > \tfrac{n}{2} \,\Big] \in \{0,1\}.$$

X_t is a function of $(\omega_0, \dots, \omega_t)$.

Lemma 6.2. *For every t and every prefix $(\omega_0, \dots, \omega_{t-1})$, $\mathbb{P}[X_t = 1 \mid \omega_0, \dots, \omega_{t-1}] \ge \frac{1}{8}$.*

Proof. The prefix determines I_t. If $m_t > n/2$, then $X_t = 1$ with conditional probability 1. Otherwise $1 \le m_t \le n/2$ (note $m_t \ge m_0 = 1$ by Lemma 1.4), and since ω_t is uniform on R_n and independent of the prefix, Lemma 4.3 applies. □

Lemma 6.3 (Enough good rounds force half-saturation)**.** *Let $L \in \mathbb{N}$ satisfy $(9/8)^L > n/2$. If ω is such that $\sum_{t<T_1} X_t(\omega) \ge L$, then $m_{T_1}(\omega) > n/2$.*

Proof. Suppose $m_{T_1} \le n/2$. By monotonicity (Lemma 1.4), $m_t \le n/2$ for all $t \le T_1$. Hence every good round $t < T_1$ satisfies $m_{t+1} \ge \frac{9}{8} m_t$, and every round satisfies $m_{t+1} \ge m_t$. By induction on the number of good rounds, $m_{T_1} \ge (9/8)^L\, m_0 = (9/8)^L > n/2$, a contradiction. □

Lemma 6.4 (Phase 1)**.** *With $T_1 = \lceil 117 \log n \rceil + 23$,*

$$\mathbb{P}\big[\, m_{T_1} \le \tfrac{n}{2} \,\big] \;\le\; \frac{1}{n}.$$

Proof. Set $L := \lceil 9 \log n \rceil + 1$; since $\log \frac{9}{8} \ge \frac{1}{9}$ (Lemma 3.4), we have $\log\big((9/8)^L\big) = L \log \frac{9}{8} \ge 9 \log n \cdot \frac{1}{9} = \log n$, hence $(9/8)^L \ge n > n/2$. By Lemma 6.3 and Corollary 5.2 (with $p = 1/8$, $s = 1/2$, justified by Lemma 6.2),

$$\mathbb{P}\big[m_{T_1} \le \tfrac{n}{2}\big] \;\le\; \mathbb{P}\Big[\textstyle\sum_{t<T_1} X_t < L\Big] \;\le\; 2^L \left(\tfrac{15}{16}\right)^{T_1}.$$

It remains to check $2^L (15/16)^{T_1} \le 1/n$, equivalently

$$T_1 \log \tfrac{16}{15} \;\ge\; L \log 2 + \log n.$$

Using $\log \frac{16}{15} \ge \frac{1}{16}$, $\log 2 < 0.6932$ and $L \le 9 \log n + 2$:

$$L \log 2 + \log n \;\le\; (9 \log n + 2) \cdot 0.6932 + \log n \;\le\; 7.239 \log n + 1.387,$$

while

$$T_1 \log \tfrac{16}{15} \;\ge\; \frac{117 \log n + 23}{16} \;\ge\; 7.312 \log n + 1.437.$$ □

Remark 6.5 (On constants)**.** *The clean statement to formalize is:* for all L, T_1 with $(9/8)^L > n/2$ and $2^L (15/16)^{T_1} \le 1/n$, we have $\mathbb{P}[m_{T_1} \le n/2] \le 1/n$ *— the numerics above merely exhibit admissible values $L = \lceil 9 \log n \rceil + 1$, $T_1 = \lceil 117 \log n \rceil + 23$; any larger T_1 also works.*

7 Saturation phase

Lemma 7.1 (Phase 2)**.** *Let $A = \{m_{T_1} > n/2\} \subseteq \Omega$ (an event determined by the first T_1 coordinates). Then for every $s \geq 0$,*

$$\mathbb{E}\big[\, U_{T_1+s}\, \mathbf{1}_A \,\big] \;\leq\; \left(\frac{2}{3}\right)^s n.$$

Proof. Induction on s. For $s = 0$: $U_{T_1} \leq n$ pointwise. For the step, condition on the first $T_1 + s$ coordinates: on A we have $m_{T_1+s} \geq m_{T_1} > n/2$ by monotonicity, so Lemma 4.4 applies to the (uniform, independent) round ω_{T_1+s} and gives, pointwise on each prefix in A,

$$\mathbb{E}\big[U_{T_1+s+1} \mid \omega_0, \ldots, \omega_{T_1+s-1}\big] \;\leq\; \tfrac{2}{3}\, U_{T_1+s}.$$

Multiplying by $\mathbf{1}_A$ (a function of the prefix), taking outer expectations, and applying the induction hypothesis finishes the proof. ☐

Corollary 7.2. *With $T_2 = \lceil 6 \log n \rceil$: $\mathbb{P}\big[\, U_T \geq 1$ and $A \,\big] \leq n\,(2/3)^{T_2} \leq 1/n$.*

Proof. U_T is $\mathbb{N}$-valued, so by Markov (Lemma 3.5) applied to $U_T \mathbf{1}_A$ and Lemma 7.1, $\mathbb{P}[U_T \mathbf{1}_A \geq 1] \leq (2/3)^{T_2} n$. And $(2/3)^{T_2} n \leq 1/n$ iff $T_2 \log \frac{3}{2} \geq 2 \log n$, which holds since $\log \frac{3}{2} \geq \frac{1}{3}$ (Lemma 3.4) and $T_2 \geq 6 \log n$. ☐

8 Proof of Theorem 2.1

With $T_1 = \lceil 117 \log n \rceil + 23$, $T_2 = \lceil 6 \log n \rceil$, $T = T_1 + T_2$:

$$\mathbb{P}[I_T \neq V] \;=\; \mathbb{P}[U_T \geq 1] \;\leq\; \mathbb{P}[\,\overline{A}\,] + \mathbb{P}[\, U_T \geq 1 \text{ and } A\,] \;\leq\; \frac{1}{n} + \frac{1}{n}$$

by Lemma 6.4 and Corollary 7.2. ☐

9 The Lean formalization

The argument above is fully formalized in the accompanying repository (library `RumorSpread`, namespace `RumorPush`); the build contains no `sorry` and the main theorem depends only on the three standard axioms (`propext`, `Classical.choice`, `Quot.sound`). Design notes:

- **Probability space.** No measure theory, no PMF, no ENNReal. `avg` is a finite sum divided by a cardinality, and `expList α T F` — the expectation of a trajectory functional F over T i.i.d. uniform draws — is defined *by recursion on* T (average out the first round, recurse). "Conditioning on the first round" is thereby a definitional unfolding, and Markov / union bounds are pointwise inequalities pushed through monotonicity of `expList`. The bridge theorem `expList_eq_avg_ofFn` (`Equivalence.lean`) proves this equals the uniform average over the product space `Fin T → R n` of all round sequences, so the model is the textbook one.
- **Process** (`Model.lean`): a round configuration is `R n := ∀ v : Fin n, {u // u ≠ v}`; one round is `step I r := I ∪ I.image (fun v => (r v).1)`; trajectories are lists of rounds consumed by `run`. Lemma 6.3 is the deterministic `pow_goodCount_mul_card_le_card_run`.
- **Lemma 4.1** (`avg_not_contacted` in `OneRound.lean`) is the only place where a probability is *computed* rather than estimated: a counting argument via an explicit equivalence to a product of subtypes and `Fintype.card_pi`.

- **Concentration**: Lemma 5.1 becomes the ten-line induction `expList_half_pow_goodCount` (`Growth.lean`); no Chernoff, no martingales, no couplings.
- **Analytic toolkit** (`Bounds.lean`): `one_add_mul_le_pow` (Bernoulli) and short inductions for Lemmas 3.2 and 3.3. `Real.log` enters only through the *definition* of L, T_1, T_2 and the numeric checks of Section 6, via Lemma 3.4 and Mathlib's `Real.log_two_lt_d9`; `Real.exp` appears only inside the proof of `numeric_A`.
- **File structure:** `Bounds.lean` (Section 3), `Prob.lean` (avg/expList), `Model.lean` (Section 1), `OneRound.lean` (Section 4), `Growth.lean` (Sections 5–6), `Saturation.lean` (Section 7), `Main.lean` (main theorem `push_informs_all_whp`), `Equivalence.lean` (faithfulness).

References

[1] A. Frieze, G. Grimmett. *The shortest-path problem for graphs with random arc-lengths.* Discrete Applied Mathematics 10 (1985) 57–77.

[2] B. Pittel. *On spreading a rumor.* SIAM J. Appl. Math. 47 (1987) 213–223.

[3] R. Karp, C. Schindelhauer, S. Shenker, B. Vöcking. *Randomized rumor spreading.* FOCS 2000.